

Branch: Mechanical Engg. (Third Year B.Tech.)
Subject with Subject Code:- Theory of Machines -II

Sem.: - V
(BTMEC504)

Marks: 60

Date:- January 2023

Time:- 3 Hrs.

Instructions:-

1. Figures to the right indicate full marks.
2. Clearly mention the main question number along with the sub questions.
3. Assume suitable data, if necessary.
4. All questions are compulsory.

Q.No.	Question	Marks	CO
1	Solve Any Two : a) A belt drive transmits 8 kW of power from a shaft rotating at 240 rpm to another shaft rotating at 160 rpm. The belt is 8 mm thick. The diameter of smaller pulley is 600 mm & the two shafts are 5 m apart. The coefficient of friction is 0.25. If the maximum stress in the belt is limited to 3 N/mm ² , find the width of the belt for i. An open belt drive, and ii. A crossed belt drive b) What are the different types of pulleys? Explain briefly. c) Derive the relation for ratio of belt tensions in a flat belt drive	06 06 06	CO1 CO1 CO1
2	Solve Any Two : a) State & derive law of gearing with neat sketch. b) Make a comparison of Cycloidal & Involute tooth forms. c) A pair of spur gears consists of a 20 teeth pinion meshing with a 120 teeth gear. The module is 4 mm. Calculate (i) the centre distance; (ii) the pitch circle diameters of the pinion and the gear; (iii) the addendum and dedendum; (iv) the tooth thickness; (v) the gear ratio.	06 06 06	CO2 CO2 CO2
3	Solve Any Two : a) What is a gear train? What are its main types (with sketches)? b) Sketch a automotive differential gear box & explain its working. c) An epicyclic gear train is shown in Fig.1. The number of teeth on A & B are 80 & 200. Determine the speed of the arm <i>a</i> . (i) If A rotates 100 rpm clockwise & B at 50 rpm counter-clockwise. (ii) If A rotates 100 rpm clockwise & B is stationary.	06 06 06	CO2 CO2 CO2

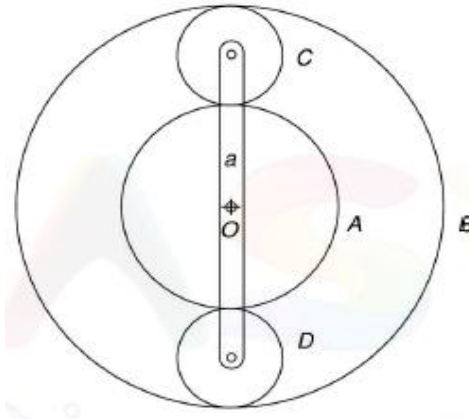


Fig.1

4 Solve Any Two:

- a) The turning moment diagram of a multi-cylinder engine is drawn with a scale of $(1 \text{ mm} = 1^\circ)$ on the abscissa and $(1 \text{ mm} = 250 \text{ N-m})$ on the ordinate. The intercepted areas between the torque developed by the engine and the mean resisting torque of the machine, taken in order from one end are $-350, +800, -600, +900, -550, +450$ and -650 mm^2 . The engine is running at a mean speed of 750 rpm and the coefficient of speed fluctuations is limited to 0.02. A rimmed flywheel made of grey cast iron FG 200 ($\rho = 7100 \text{ kg/m}^3$) is provided. The spokes, hub and shaft are assumed to contribute 10% of the required moment of inertia. The rim has rectangular cross-section and the ratio of width to thickness is 1.5. Determine the dimensions of the rim.
- b) Explain the terms sensitiveness, hunting, stability, isochronisms & effort of governor relating to the governors.
- c) Sketch a Hartnell governor. Describe its function & deduce a relation to find the stiffness of the spring.

06 CO4

5 Solve Any Two:

- a) A ship is propelled by turbine rotor having a mass of 6 tonnes & a speed of 2400 rpm. The direction of rotation of the rotor is clockwise when viewed from stern. The radius of gyration of the rotor is 450 mm. Determine the gyroscopic couple & its effect when
- The ship steers to the left in a curve of 60 m radius at a speed of 18 knots ($1 \text{ knot} = 1860 \text{ m/h}$)
 - The ship pitches 7.5 degrees above & 7.5 degrees below the normal position & the bow is descending with its maximum velocity; the pitching motion is simple harmonic with a periodic time of 18 seconds.
 - The ship rolls & at the instant, its angular velocity is 0.035 rad/s counter clockwise when viewed from stern
 - Also find the maximum angular acceleration during pitching
- b) Explain in what way the gyroscopic couple affects the motion an aircraft & naval ship?
- c) How do the effects of gyroscopic couple & of centrifugal force make the rider of a two wheeler tilt on one side? Derive a relation for angle of heel.

06 CO5

06 CO5

06 CO5

6 Solve Any Two:

- a) Explain the whirling of vertical shaft carrying a single rotor without damping & derive the equation for deflection of the shaft.

06 CO6

b) Determine the natural frequency of the system shown in Fig.2.

06 CO6

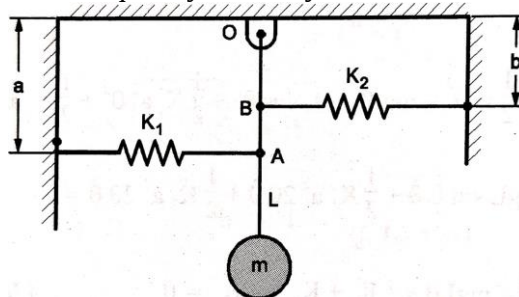


Fig.2.

c) In a forced vibratory system a body having 2 kg mass vibrates in a viscous fluid. The harmonic exciting force of 20 N acting on the mass result in a resonance amplitude of 15 mm with a period of 0.15 sec. Determine the damping coefficient of viscous fluid.

06 CO6

If the system is excited by the same harmonic force but at a frequency of 5 cycles/sec. what will be the amplitudes of forced vibration with & without damper?
